13.8 LAB: Digital forensics (Walkthrough)

Introduction

Digital forensics is a branch of forensic science focused on recovering, collecting, analyzing, and examining data found in devices. Different forensic techniques include recovering deleting files, analyzing browser activity, and examining metadata.

Conducting forensic analysis first requires a complete copy of a device. The copy must be bit-for-bit to ensure accuracy. The Linux command **dd** is used to make a bit-for-bit copy.

FTK Imager (Forensics Toolkit) is a Windows tool that copies a system similar to dd. The toolkit can also analyze emails, decrypt files, and detect malware. **E01** is a forensic image file for a software application called EnCase. Compared to dd images, E01 takes up less disk space and contains metadata.

Learning Objectives

- Create a forensics copy of a system
- View a forensics copy of a system using Autopsy
- Recover and analyze deleted files
- Generate a forensics report

Devices

The lab contains one virtual machine:

- **ZYWIN01** (Windows Server 2022)

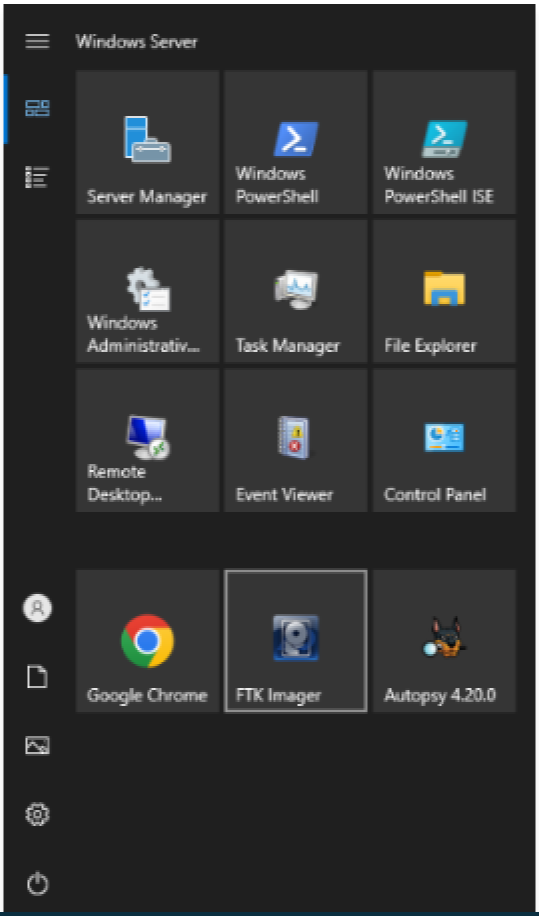
Tasks

Task 1: Convert a dd image to E01 using FTK Imager
Task 2: Use Autopsy to analyze the file system
Task 3: Generate a report with Autopsy

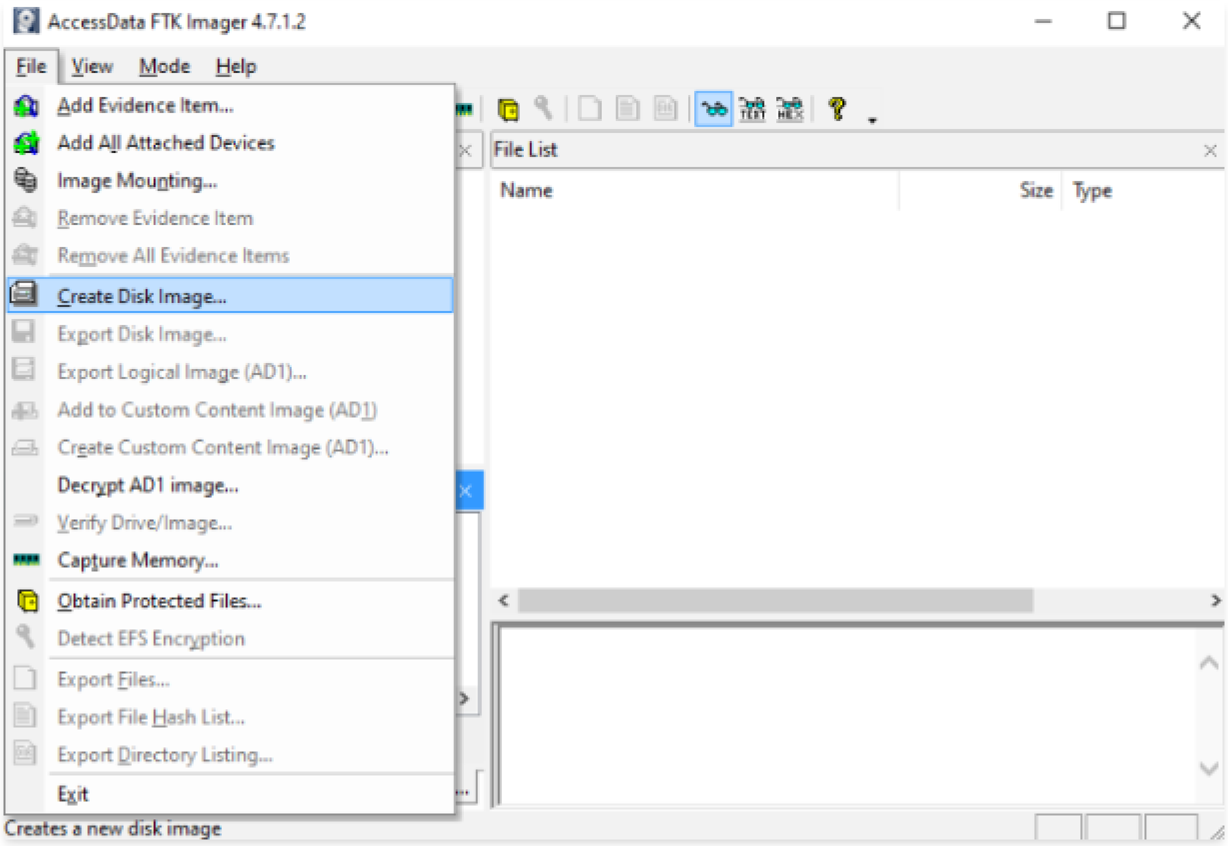
Task 1: Convert a dd image to E01 using FTK Imager

1. Select **ZYWIN01** from the server drop-down menu in the right pane and click **Connect to VM**.

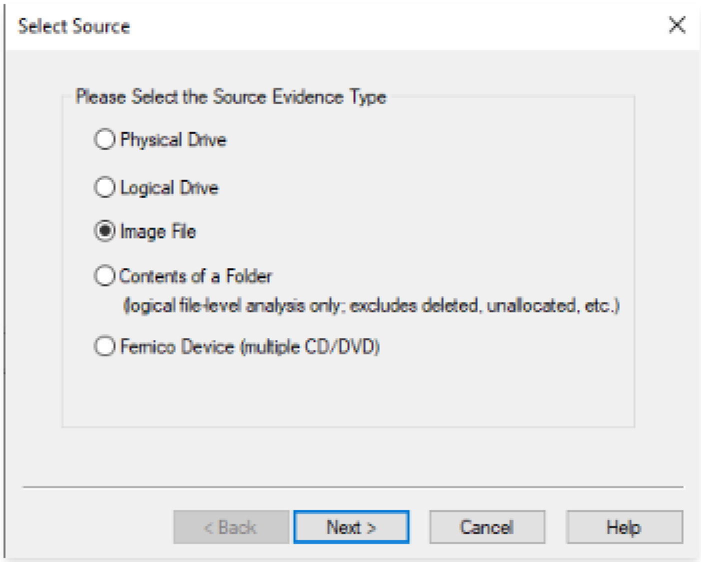
2. Select the **Windows Start Menu** at the bottom left. Select **FTK Imager**.



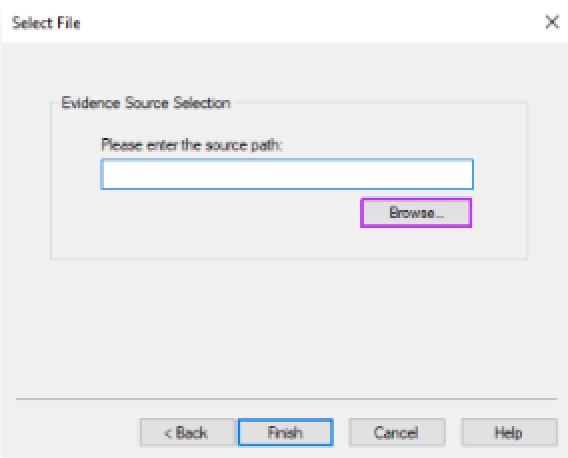
3. Select **File → Create Disk Image**.



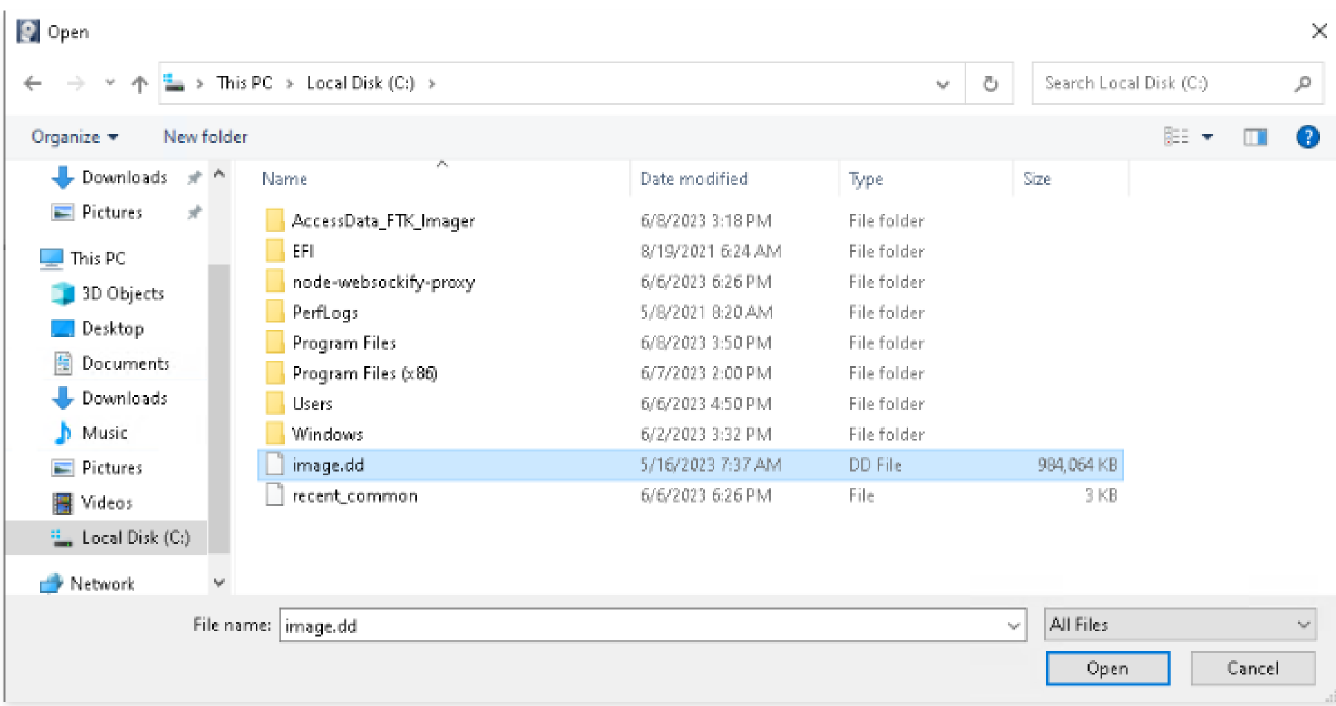
4. Select **Image File** and then **Next**.



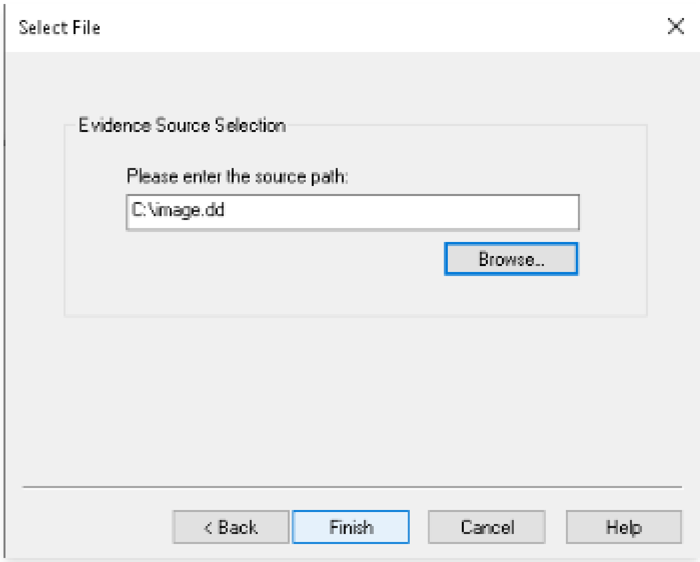
5. Click **Browse**.



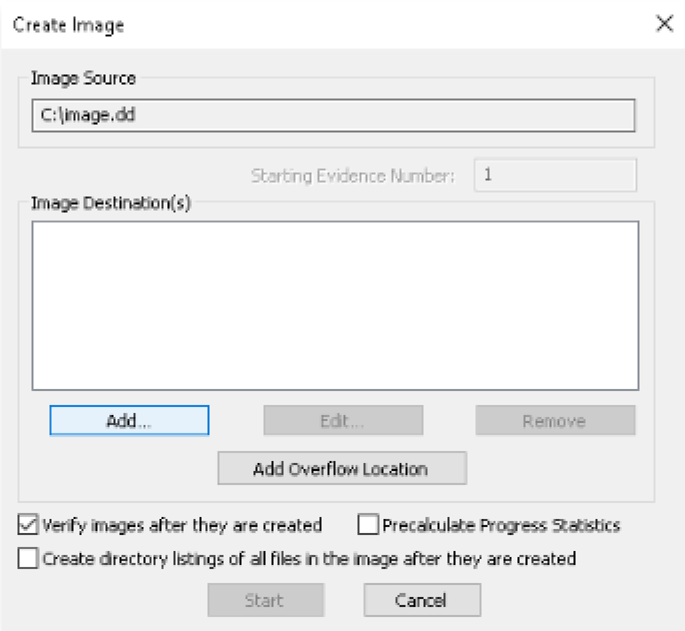
6. Go to **This PC → Local Disk (C:)**. Select **image.dd** and click **Open**.



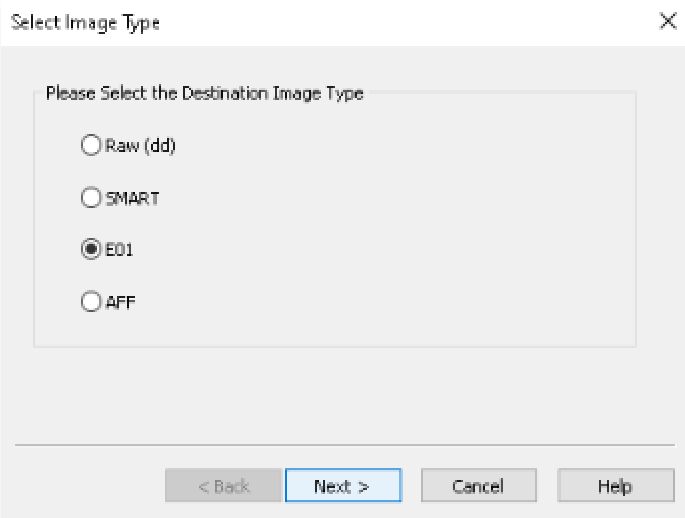
7. Click **Finish**.



8. On **Create Image**, Click **Add**.



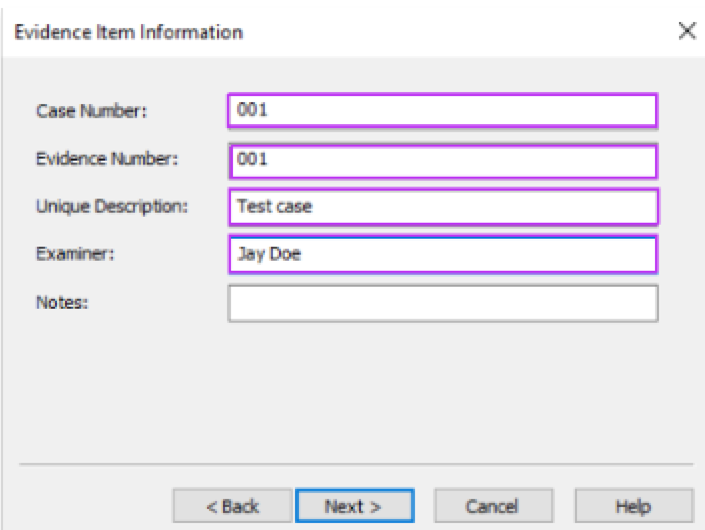
9. Choose **E01** to select the destination image type. Click **Next**.



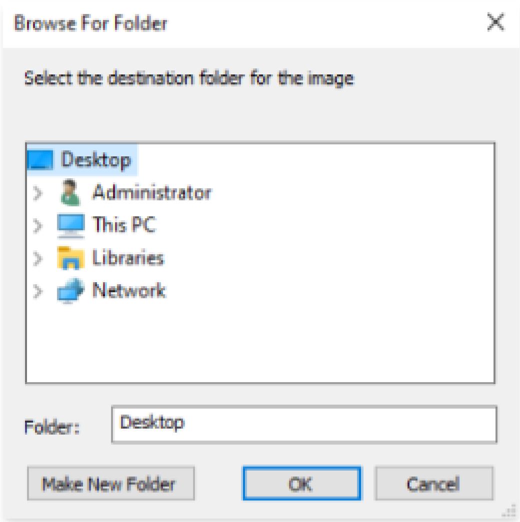
10. Enter the following information:

- **Case Number**: *001*
- **Evidence Number**: *001*
- **Unique Description**: *Test case*
- **Examiner**: *Jay Doe*

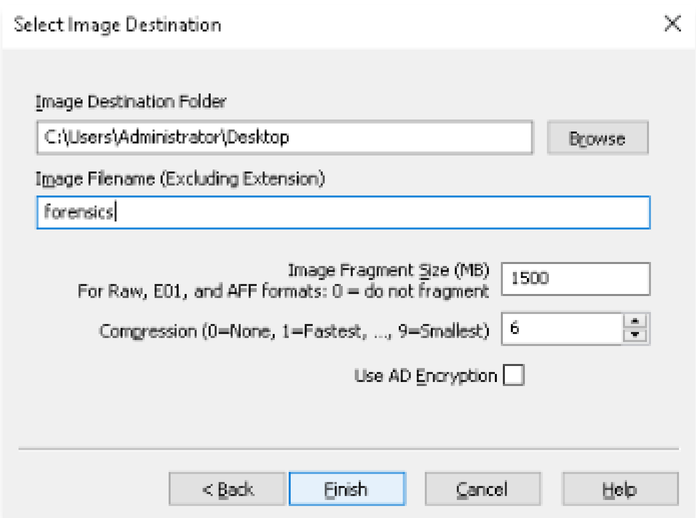
Click **Next**.



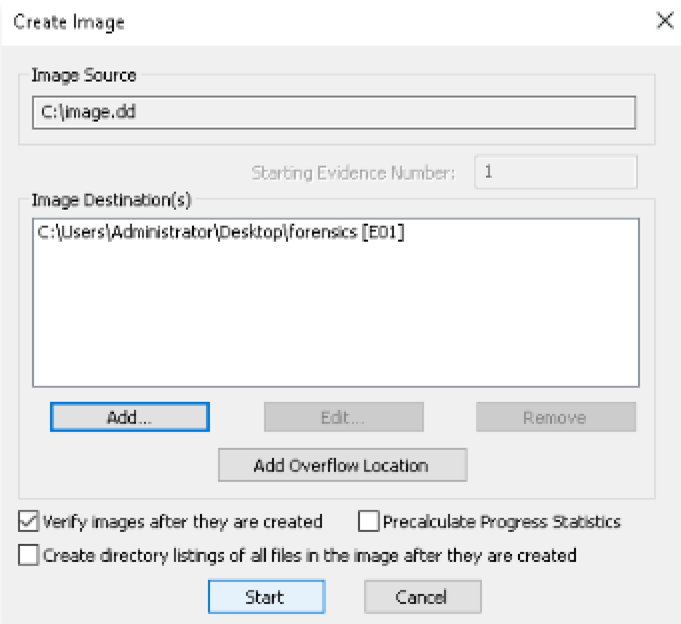
11. For image destination, click **Browse**. Select **Desktop** and click **OK**.



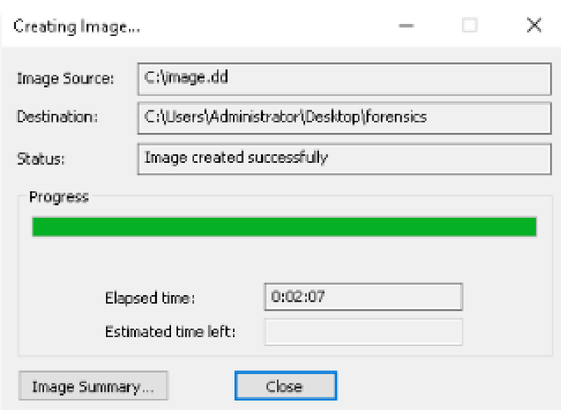
12. Enter **forensics** for **Image Filename** and click **Finish**.



13. Click **Start**.



14. Once the image is created, click **Close** on the **Drive/Image Verify Results** and **Creating Image...** windows. (Note: Image creation may take a few minutes to complete.)



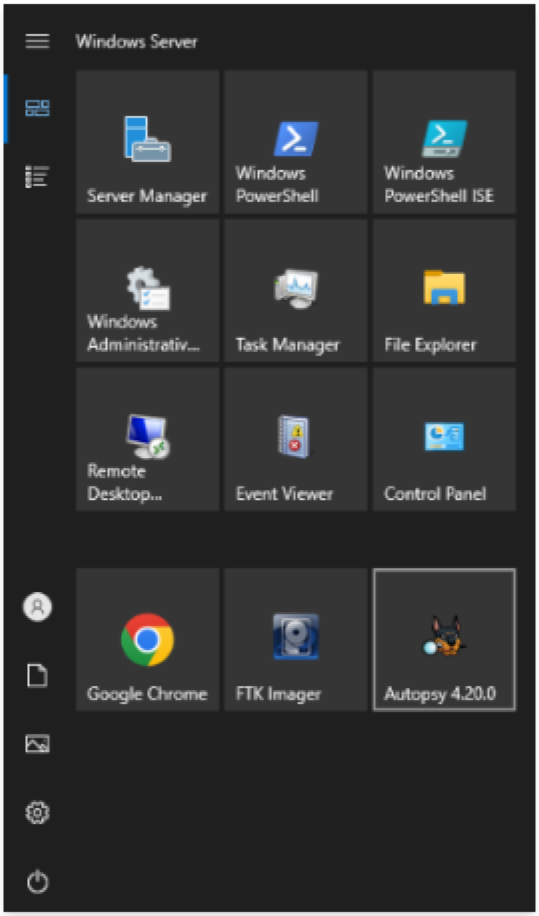
Points

0/3 E01 image created.

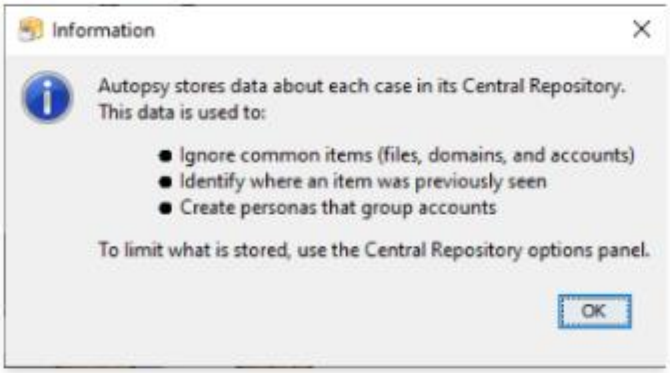
15. **Close** FTK Imager.

Task 2: Use Autopsy to analyze the file system

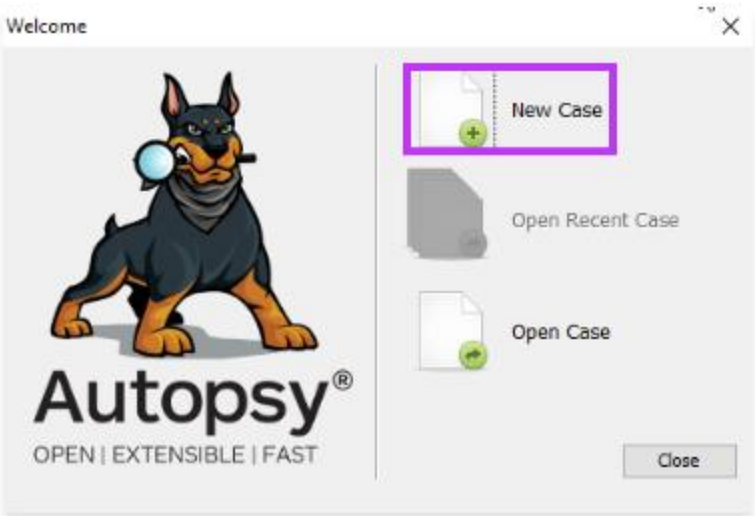
1. Open the **Windows Start Menu**. Click **Autopsy 4.20.0**.



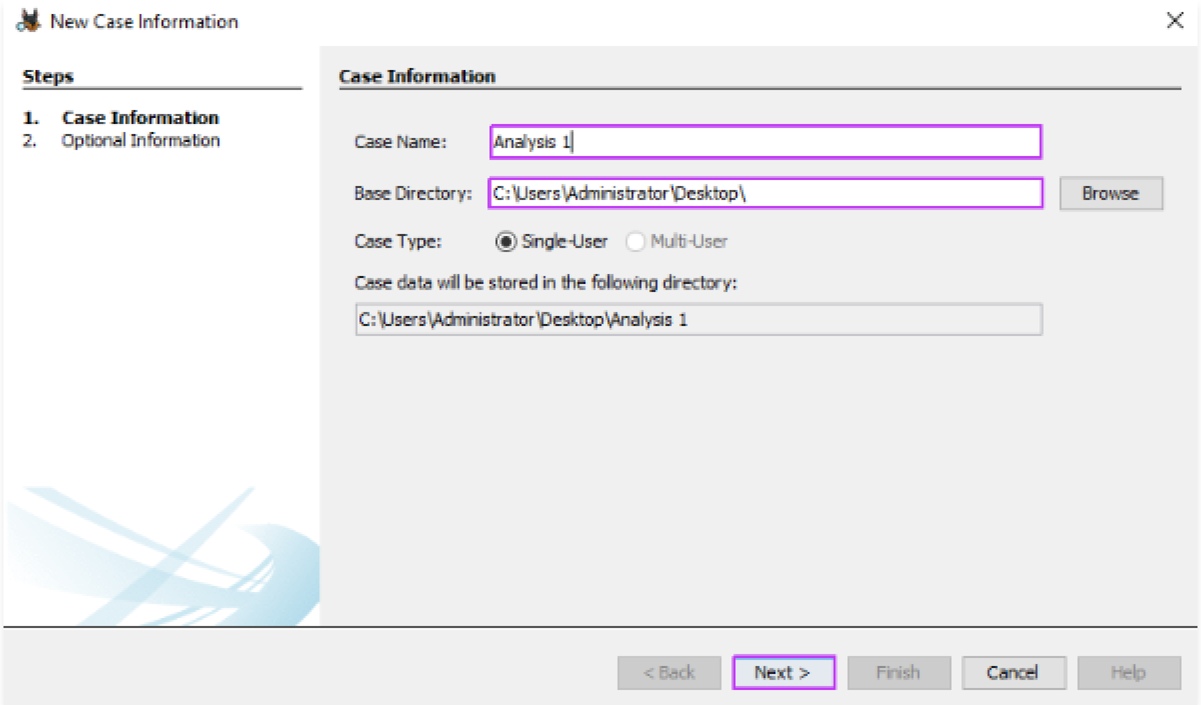
2. Click **OK**.



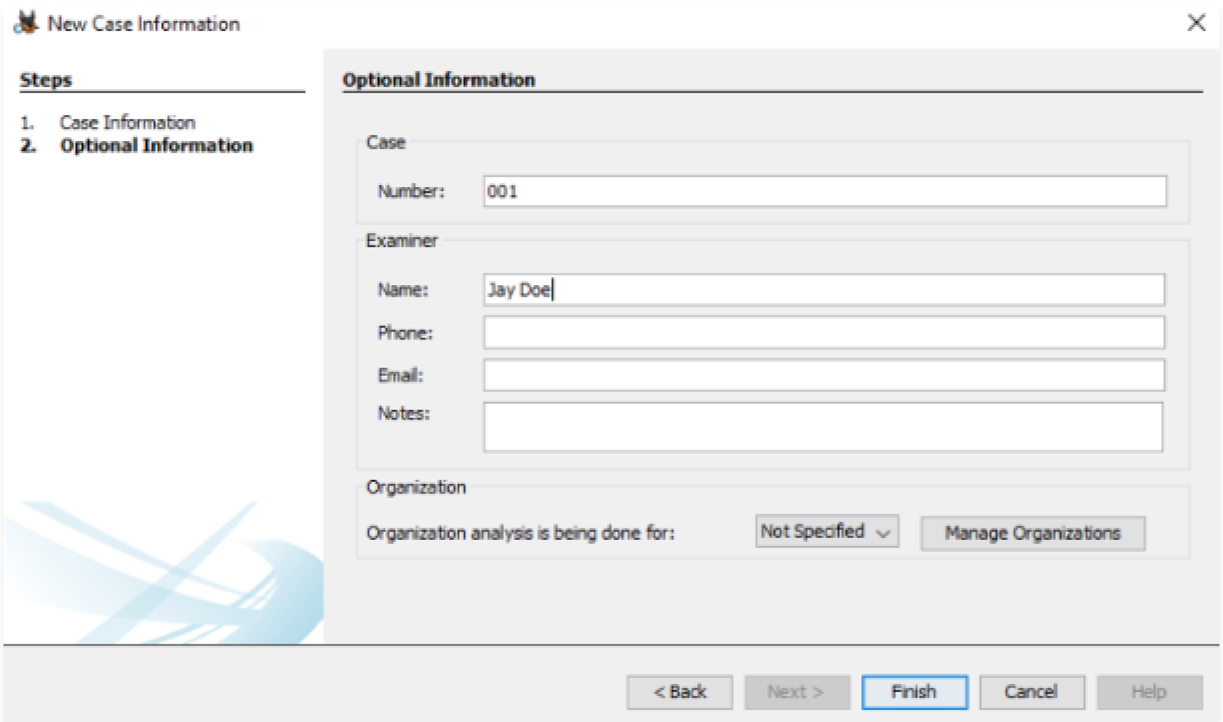
3. On the welcome screen, select **New Case**.



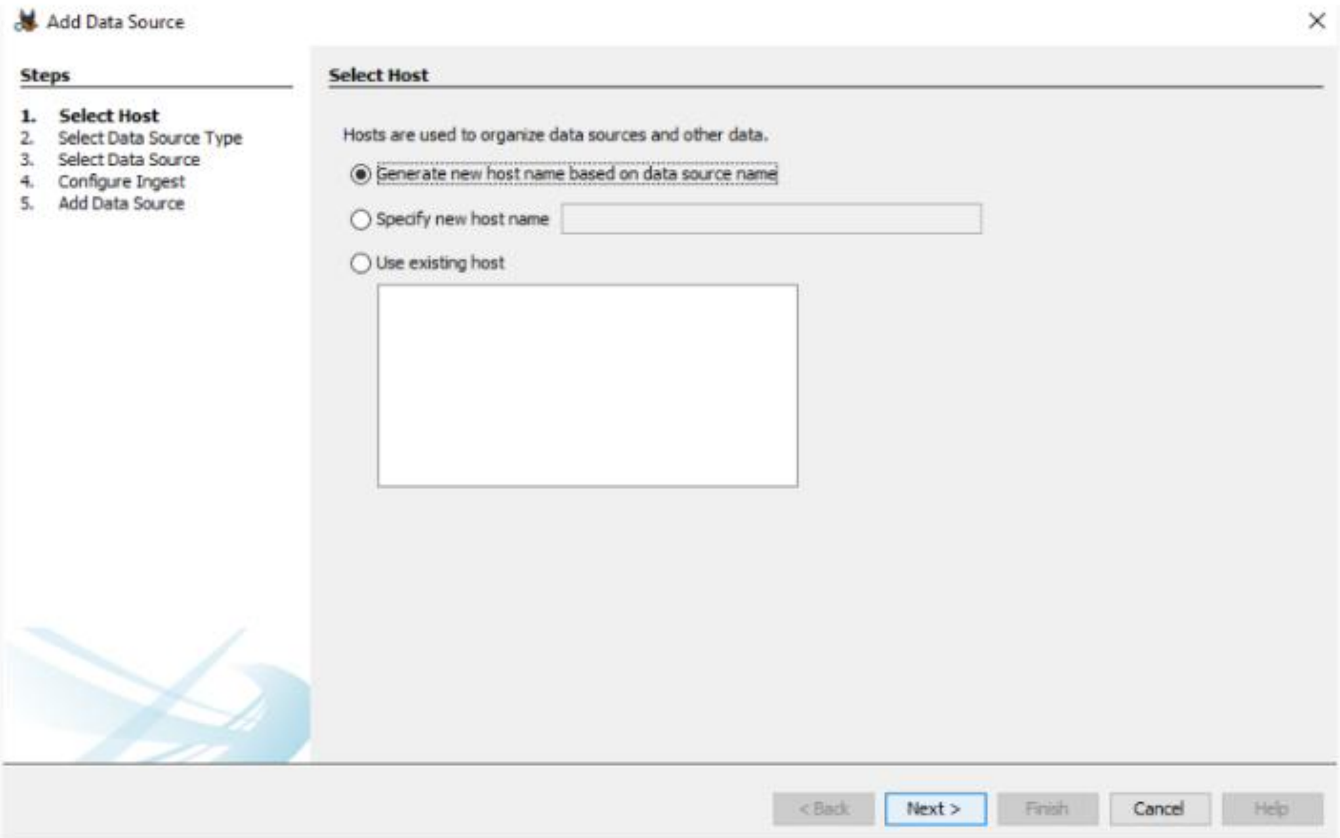
4. Enter **Analysis 1** for **Case Name**. For Base Directory, browse to select **Desktop**. Click **Next**.



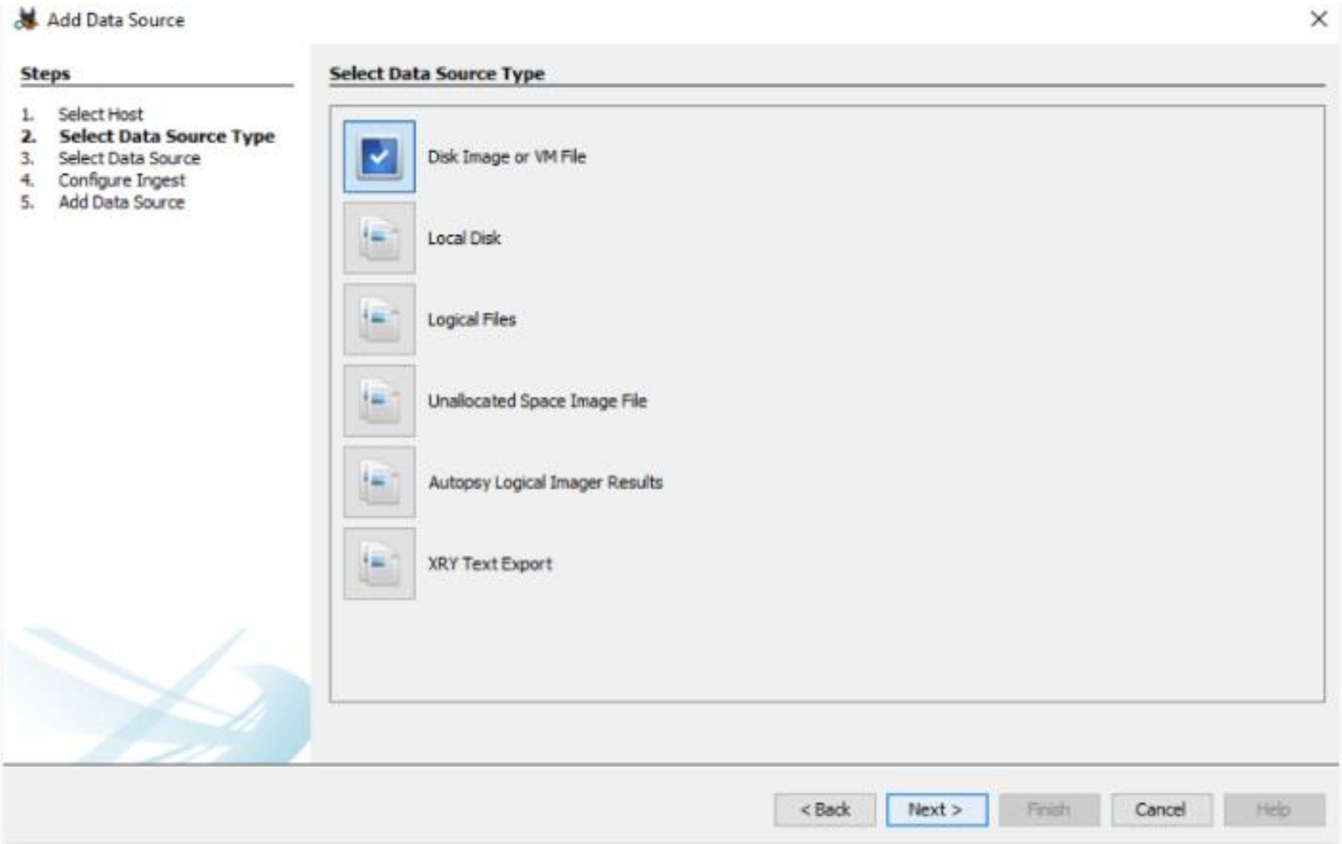
5. For **Number**, enter **001** . For **Name**, enter **Jay Doe**. Click **Finish**.



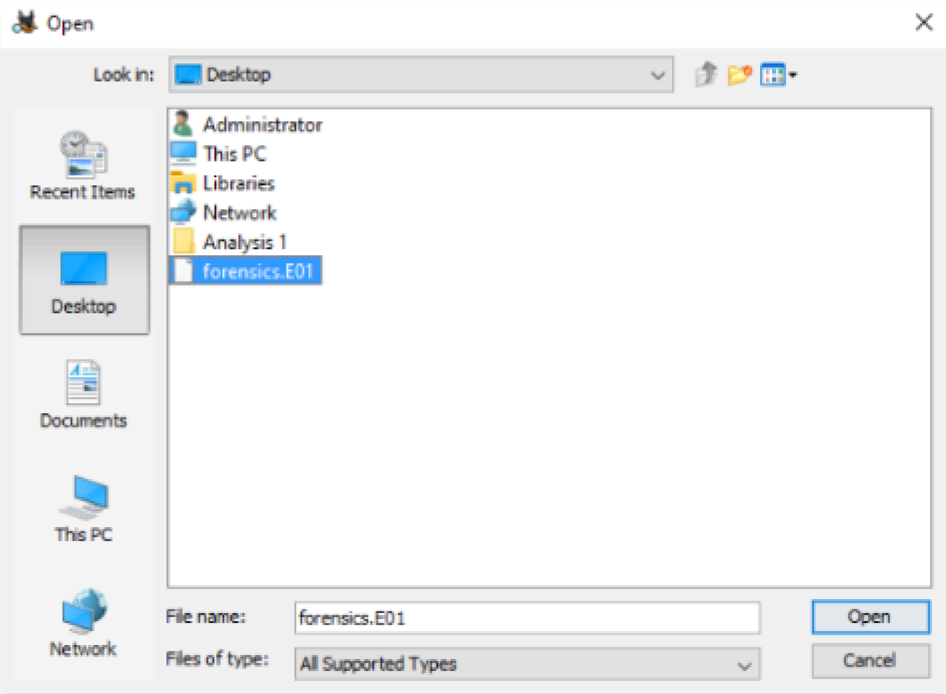
6. Select **Generate new host name based on data source name**. Click **Next**.



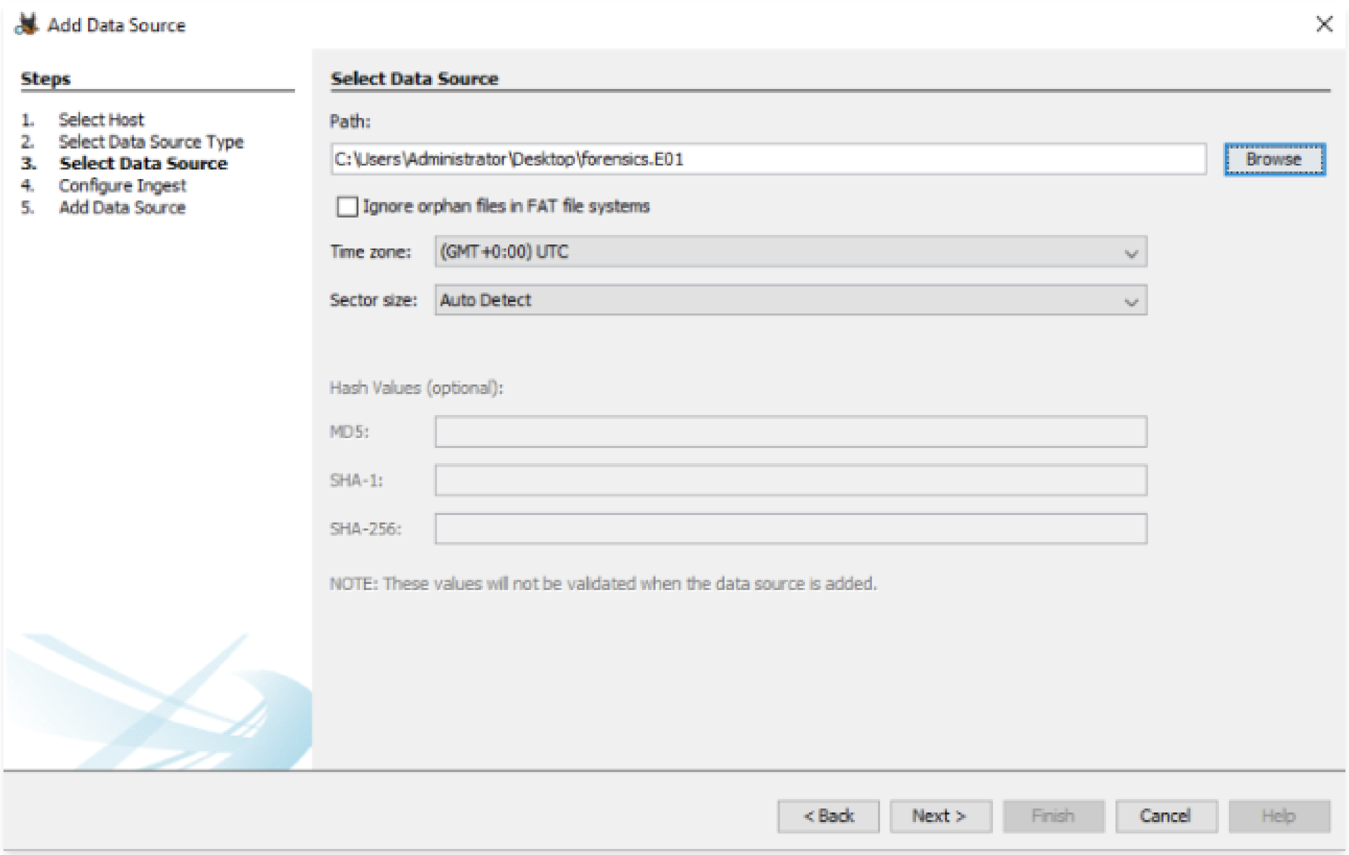
7. Select **Disk Image or VM File**. Click **Next**.



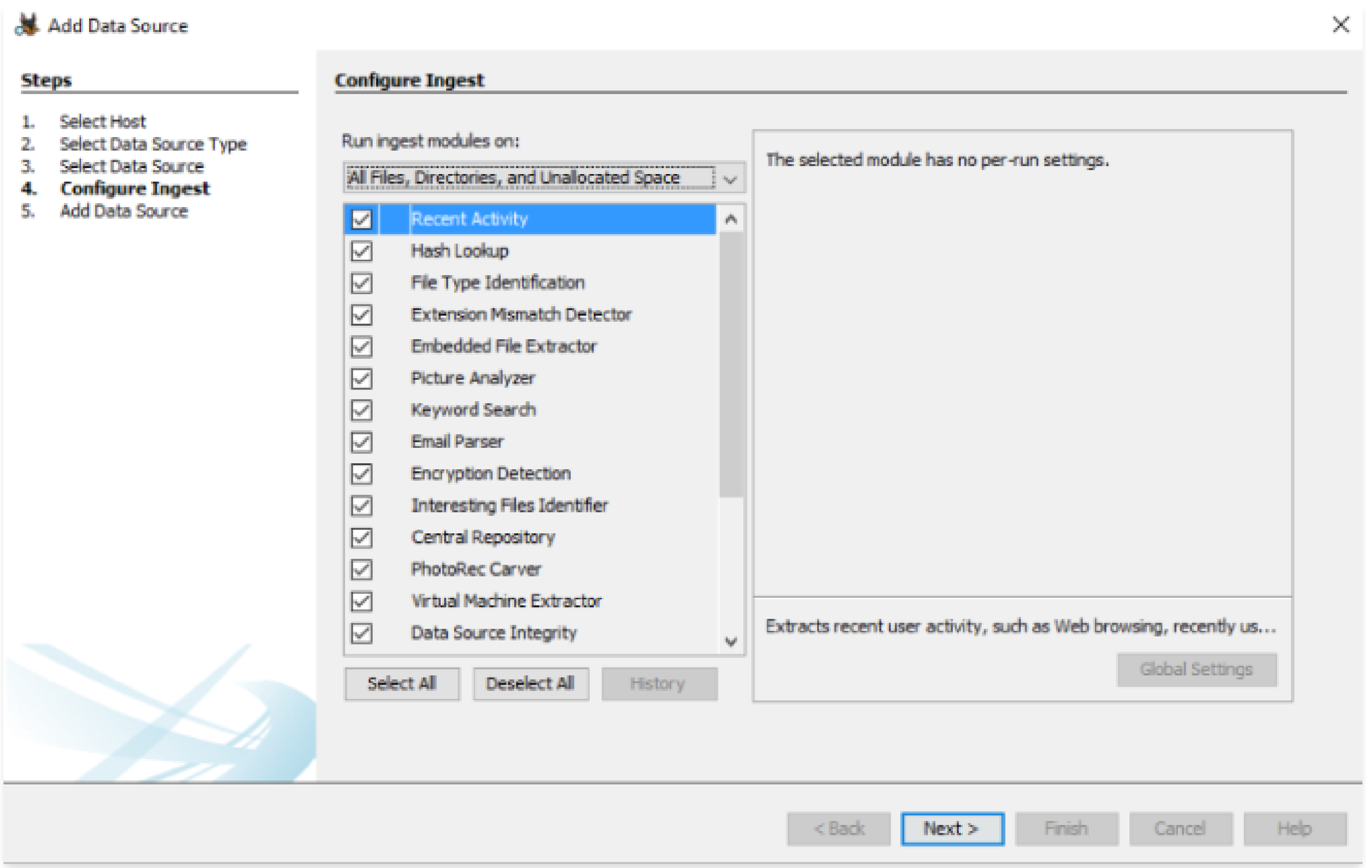
8. For the data source **Path**, select **Browse**. Select **Desktop** in the left pane and open **forensics.E01**.



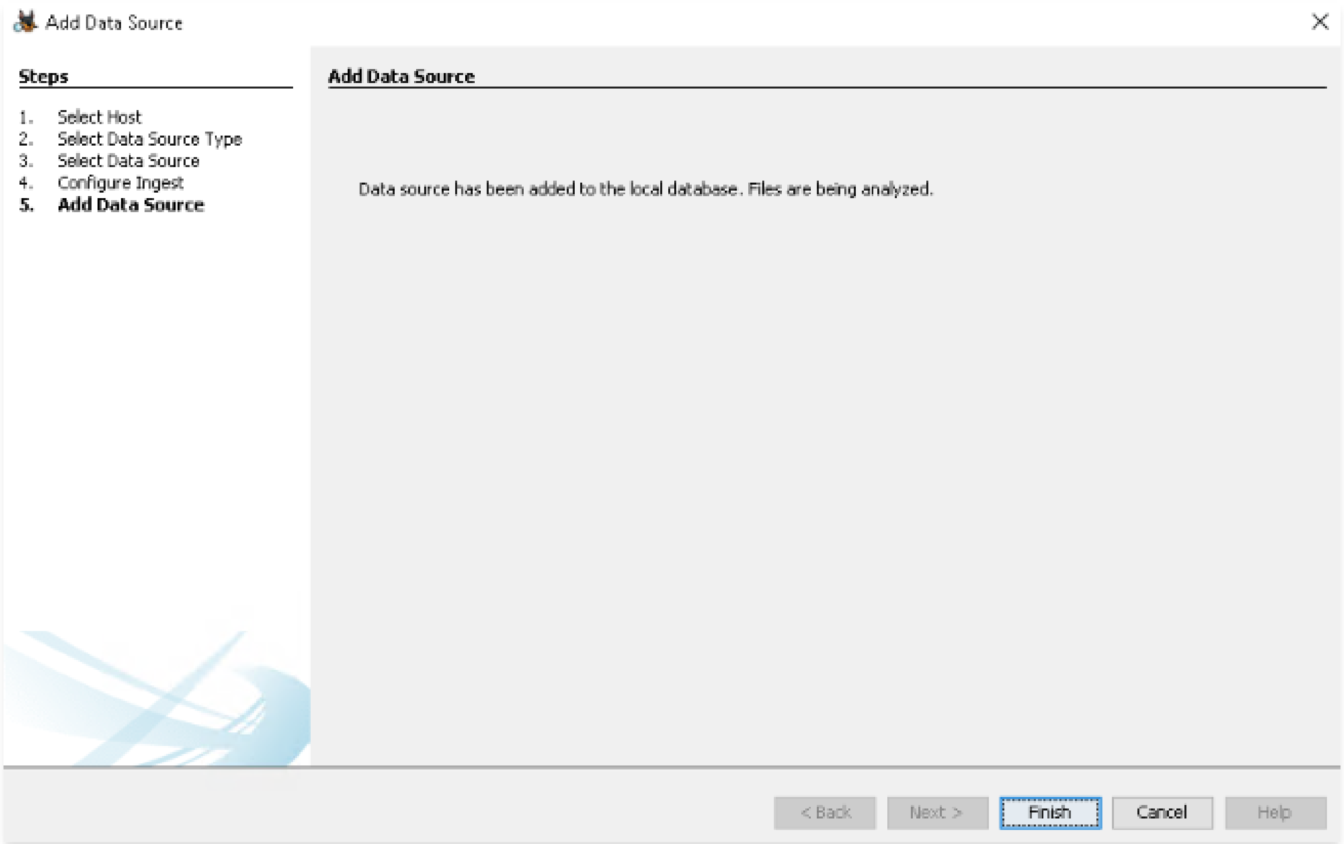
9. Click **Next**.


Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source
Path:
C:\Users\Administrator\Desktop\forensics.E01
Browse
Ignore orphan files in FAT file systems
Time zone:
(GMT+0:00) UTC
Sector size:
Auto Detect
Hash Values (optional):
MD5:
SHA-1:
SHA-256:
NOTE: These values will not be validated when the data source is added.
< Back
Next >
Finish
Cancel
Help

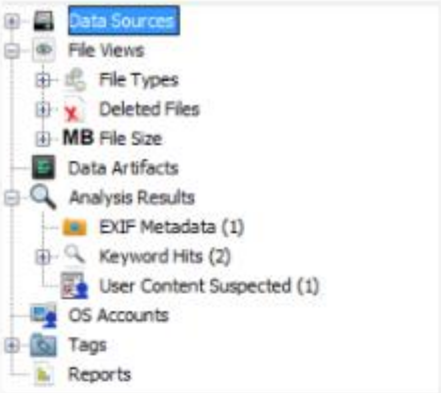
10. Click **Next**.



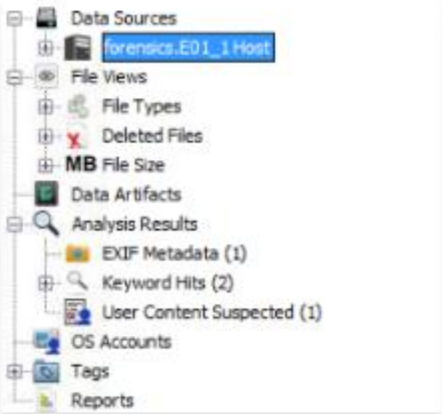
11. Click **Finish**.



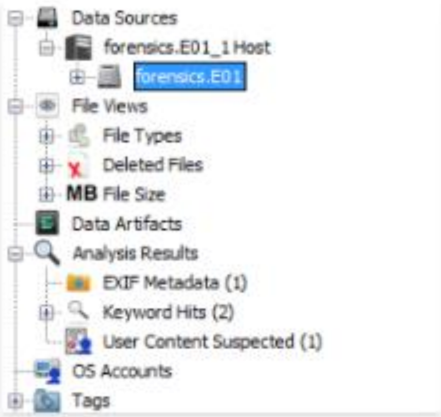
12. Expand **Data Sources** in the left pane.



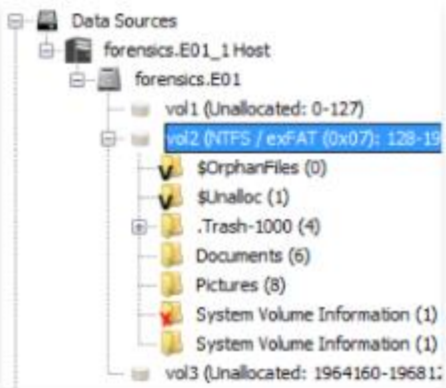
13. Click **forensics.E01_1 Host**.



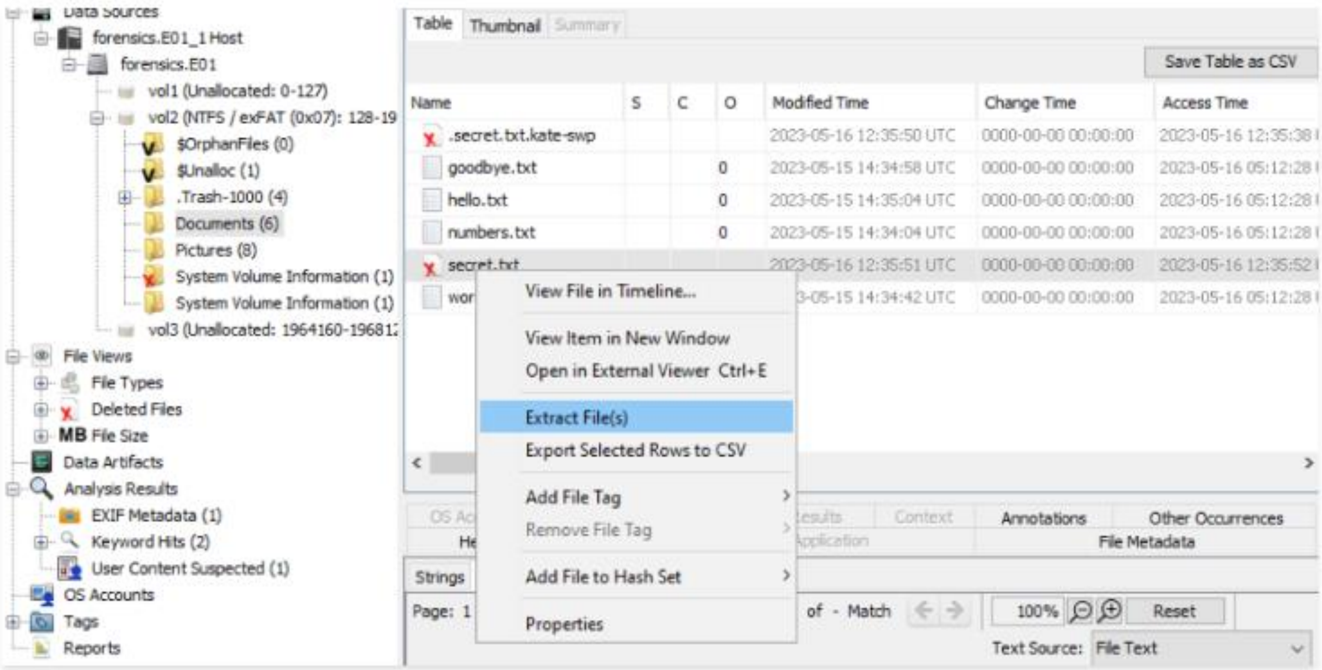
14. Click **forensics.E01**.



15. Click **vol2**.

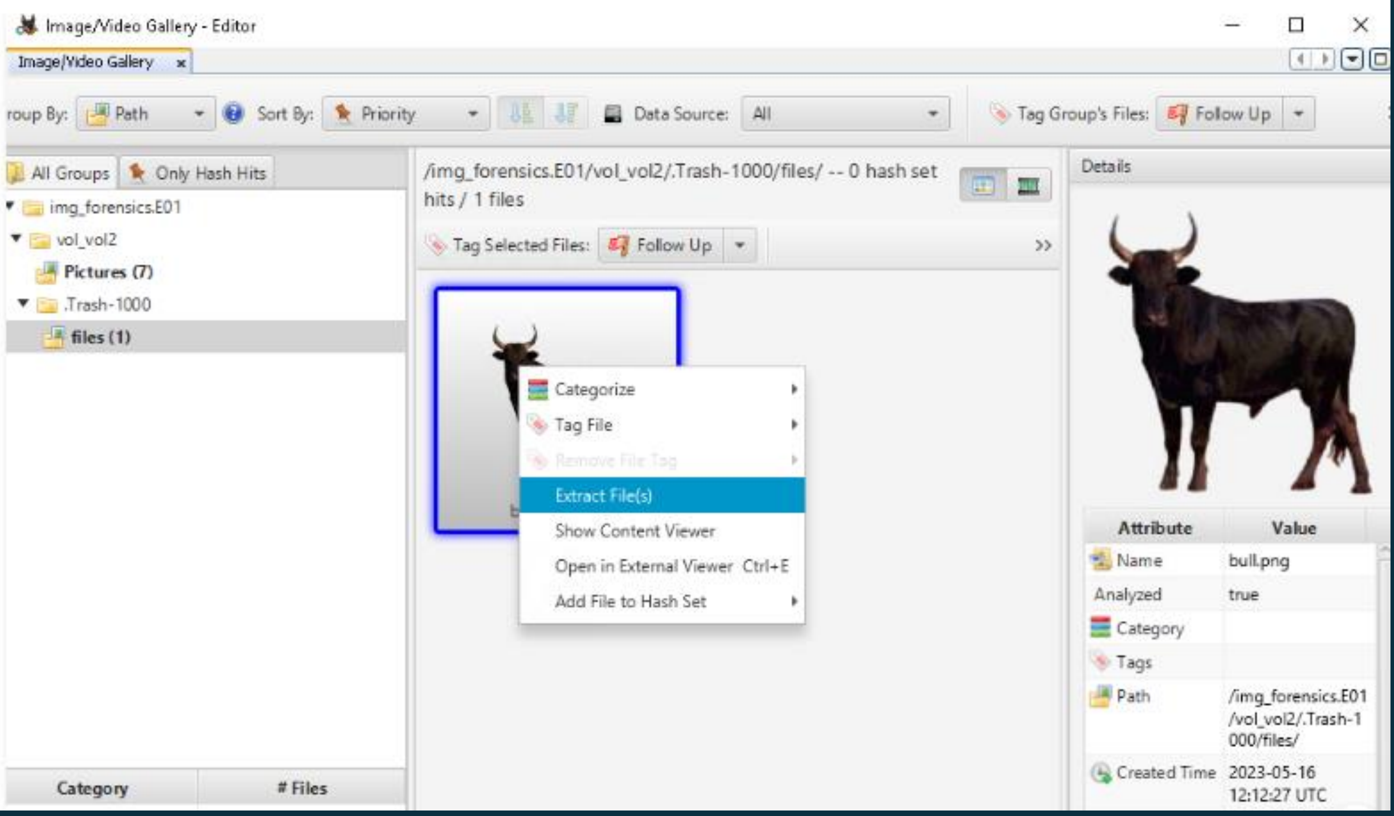


16. Select **Documents**. Right click **secret.txt** and select **Extract File(s)**. Save to desktop as **secret.txt**.

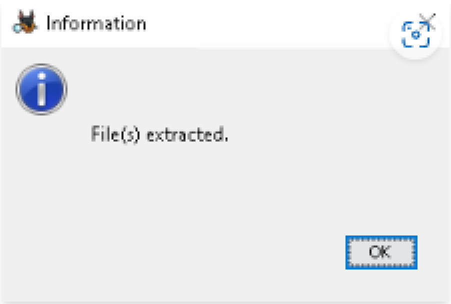


17. Click Images/Videos at the top.

18. Select **img_forensics.E01 → vol_vol2 → .Trash-1000 →** files. Right click **bull.png** and **Extract File(s)**. Save to desktop as **bull.png**.



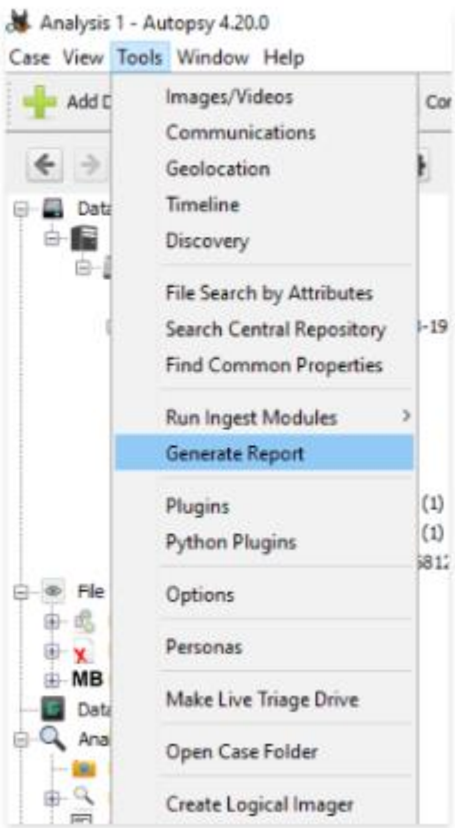
19. Click **OK** on the popup.



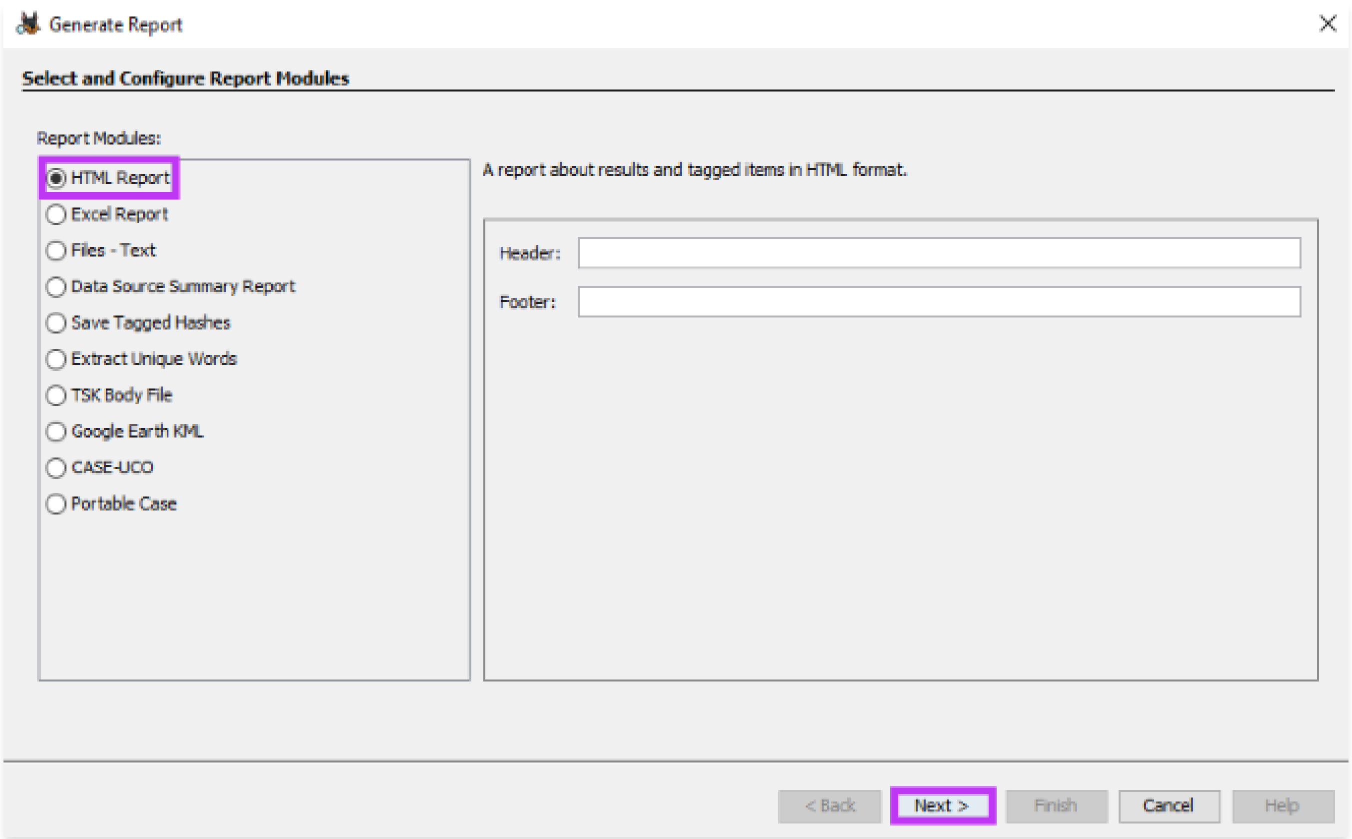
20. Close the **Image/Video Gallery - Editor**.

Task 3: Generate a report with Autopsy

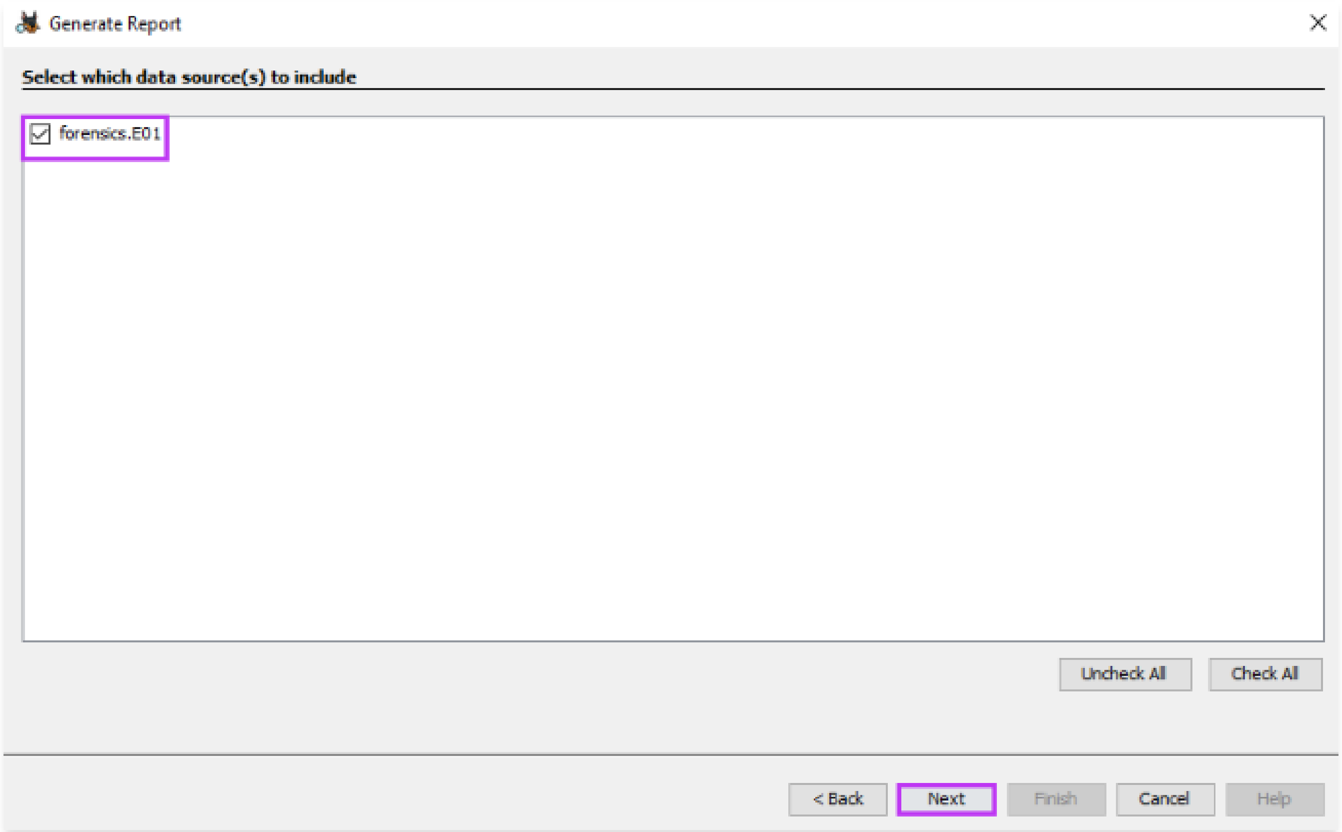
1. At the top, select **Tools** and then **Generate Report**.



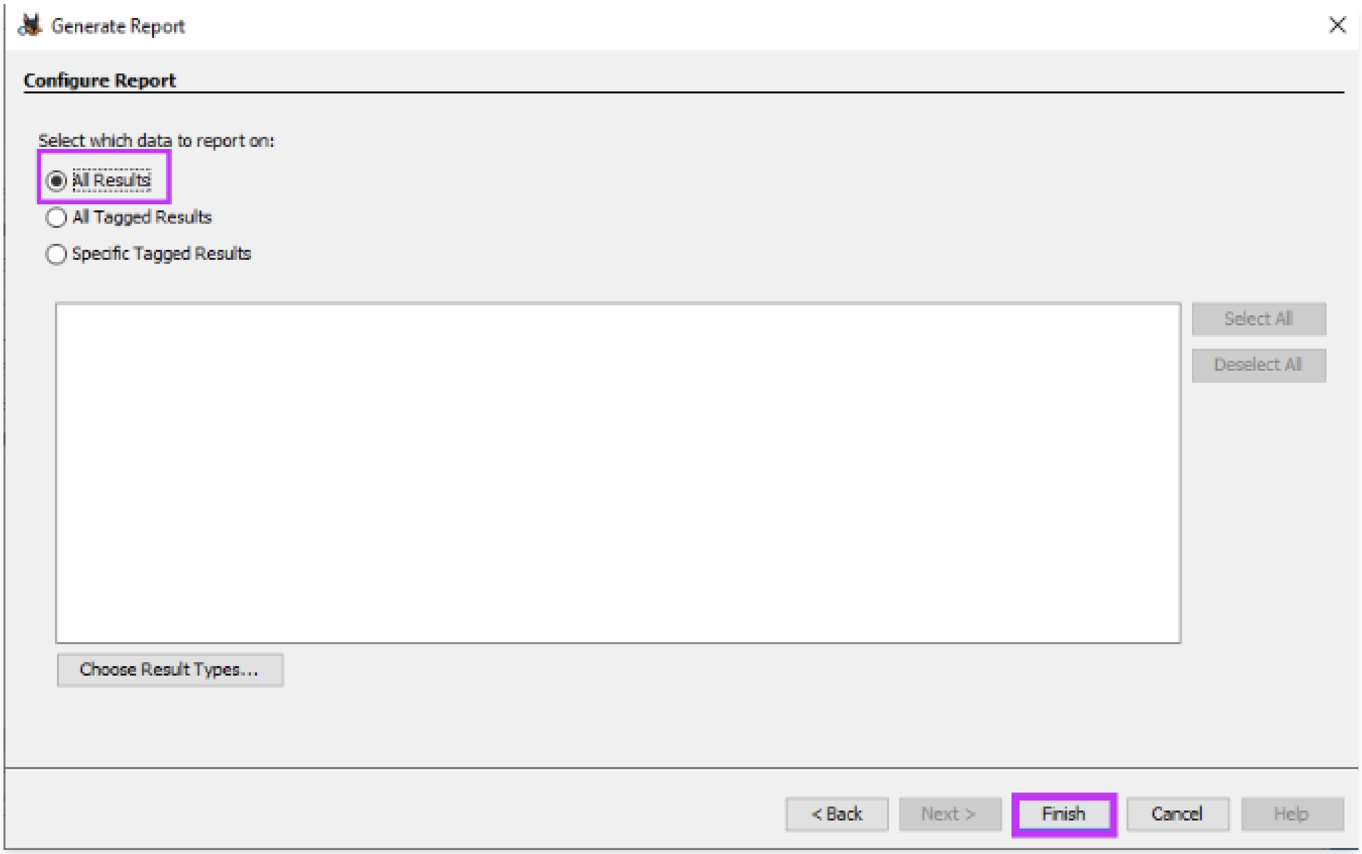
2. Select **HTML report**. Click **Next**.



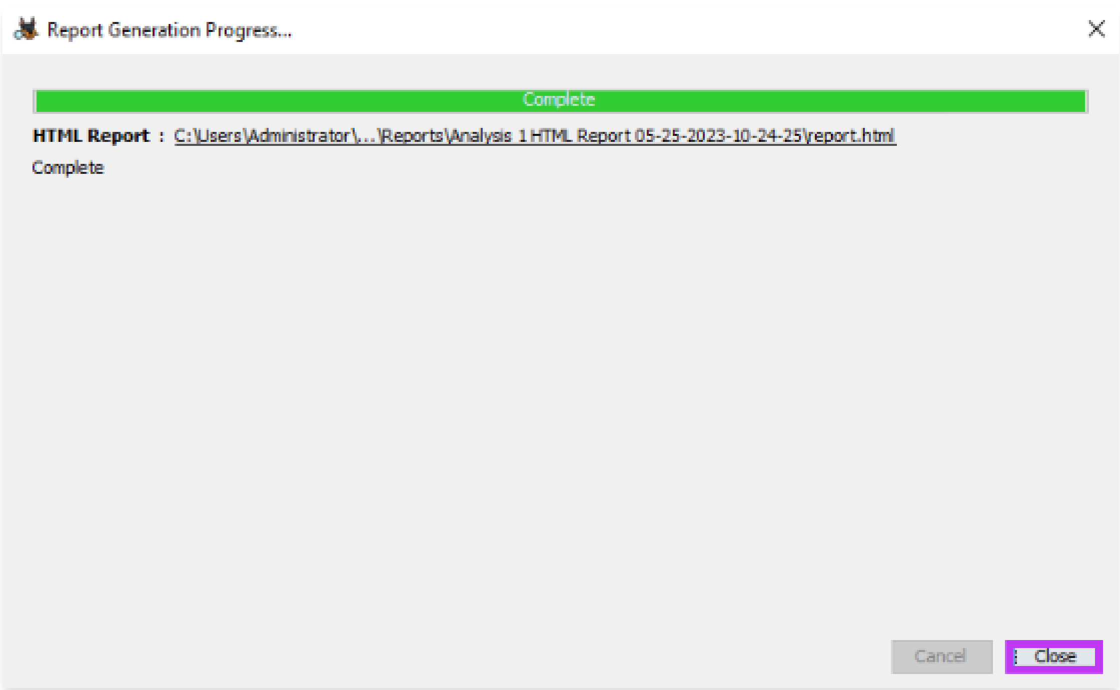
3. Check **forensics.E01**. Click **Next**.



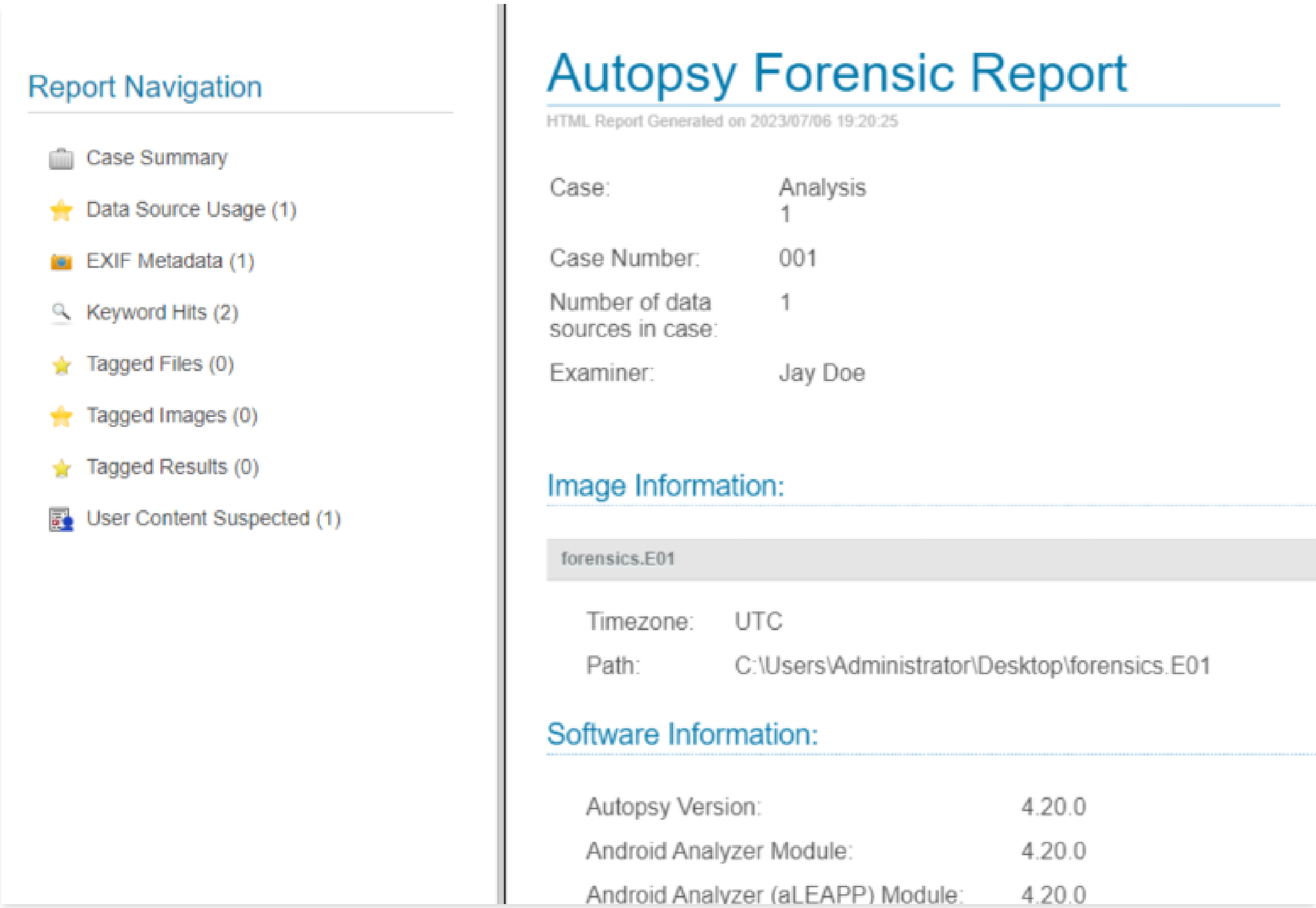
4. Select **All Results**. Click **Finish**.



5. Click **Close**.



6. Go to the location the report is saved in. Double click the report.html link to view the autopsy report.



7. The lab is now complete. To end the lab, click **Delete VM** in the right pane.